

ETHAN KNOX

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SUMMARY

Applied mathematician completing an M.S. in Applied Mathematics at UIUC (May 2026), transitioning into actuarial science with SOA Exam P scheduled for July 2026. Strong foundation in probability, statistics, and stochastic processes, backed by extensive scientific-computing experience in Python and GPU-accelerated numerical methods. Particular strength in translating complex mathematical formulations into validated, computable implementations. Seeking an actuarial analyst or development-program role.

ACTUARIAL EXAMS

SOA Exam P (Probability) — Sitting July 2026

EDUCATION

University of Illinois — Urbana-Champaign

M.S. in Applied Mathematics

Champaign, IL
Aug 2024 – May 2026

Purdue University

B.S. in Mathematics (Minor: Psychology) • B.S. in Physics (Minor: Astronomy)

West Lafayette, IN
Jun 2017 – May 2021

Honors: Dean's List; Semester Honors; Provost Summer Scholarship; Frank O'Bannon Honors & Accelerated Scholarships.

EXPERIENCE

Independent Researcher — Quantitative & Computational Modeling

May 2023 – Present
Champaign, IL

Self-directed

- Building a **GPU-accelerated interest-rate and capital-modeling engine**: implemented and numerically validated a single-factor Hull-White model with Monte Carlo simulation against closed-form benchmarks (bond options, caplet pricing).
- Developed **Nulltracer**, a GPU-accelerated ray tracer for the Kerr-Newman spacetime (Python / CuPy / CUDA) implementing five high-order ODE integrators — an exercise in numerical integration and performance optimization.

Graduate Research Assistant — Kirr Research Group, Dept. of Mathematics

Aug 2025 – Dec 2025
Champaign, IL

University of Illinois — Urbana-Champaign

- Co-authored a research report on the existence, uniqueness, and stability of multi-peak ground-state solutions of the nonlinear Schrödinger equation.
- Reduced an infinite-dimensional model to a tractable finite-dimensional system using dimension-reduction techniques — an approach conceptually adjacent to actuarial reserving and asset-liability modeling.
- Applied probabilistic and asymptotic analysis: derived explicit admissibility constraints and resolved symmetry-driven obstructions via perturbation methods.

Undergraduate Research Assistant — Iyer-Biswas Lab

Aug 2020 – Dec 2020
West Lafayette, IN

Purdue University

- Built custom Python pipelines to extract quantitative measurements from large experimental datasets, contributing to published research on stochastic single-cell behavior.
- Applied stochastic-process analysis to single-cell biophysics; presented progress and edge-case analysis in weekly all-staff meetings.

SKILLS

Quantitative & Actuarial: Probability & statistics, stochastic processes, Monte Carlo simulation, numerical methods, partial differential equations, linear algebra, optimization, dynamical systems

Programming & Tools: Python (NumPy, SciPy, Pandas, SymPy, Matplotlib, CuPy), C/C++, CUDA / GPU computing, Git, LaTeX, Linux / Bash

ADDITIONAL

Eagle Scout, Boy Scouts of America • Brazilian Jiu-Jitsu practitioner